



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

COURSE MATERIAL

Academic year: 2026-2027

Semester: I

**Course Code & Name: 25ESC231 & Introduction to
Electronics and
Communication Engineering**

Course Coordinator: SHASHIKIRAN S

Approved by:

Dr. Aravinda K HOD-ECE

INTRODUCTION TO ELECTRONICS AND COMMUNICATIONS ENGINEERING													
Course Code	25ESC131/231							CIE Marks		50			
L:T:P:S	3:0:0:0							SEE Marks		50			
Hrs / Week	4							Total Marks		100			
Credits	03							Exam Hours		03			
Course outcomes: At the end of the course, the student will be able to:													
CO1	Explain the properties of semiconductors, the operation of PN junction diodes, and their applications in rectifiers and power supply circuits												
CO2	Apply the principles of semiconductor devices for the construction, characteristics, and applications of BJT and FET												
CO3	Analyze the operation of operational amplifiers and oscillators for different applications												
CO4	Illustrate the fundamentals of communication systems, types of modulation, and basic receiver principles												
CO5	Apply the principles of digital logic, including number systems, Boolean algebra, and logic gates, to solve basic digital design problems												
CO6	Engage in independent learning as a member of a team, and use ICT for effective presentation of the study related to electronic systems												
Mapping of Course Outcomes to Program Outcomes and Program Specific Outcomes:													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-
CO6	3	3	2	1	2	-	1	2	2	-	2	-	-
MODULE-1	INTRODUCTION TO SEMICONDUCTOR: DIODE AND APPLICATIONS							25ESC131/231.1 25ESC131/231.6			8 Hours		
Introduction to materials used in electronics, P-N Junction diode – operation, characteristics, Applications: Half-Wave rectifier, Center-tapped Full Wave Rectifier, Bridge Rectifier, Power supply filters, Voltage regulators.													
Self-study		Study the basic structure of PV solar cell.											
Text Book		Textbook 1: 1.2, 2.1, 2.2, 2.4, 2.5, 2.6											
MODULE-2	TRANSISTORS: BJT AND FET							25ESC131/231.2 25ESC131/231.6			8 Hours		
Bipolar Junction Transistor: BJT Structure, BJT Operation, Characteristics and parameters, BJT as an Amplifier, BJT as a switch. Biasing, Need for Biasing, DC operating point (Q-point). JFET: JFET Structure, JFET Operation, JFET Characteristics													
Self-study		Phototransistor and its operation											
Text Book		Textbook 1: 4.1,4.2,4.3,4.4, 4.5, 5.1, 8.1											
MODULE-3	OPERATIONAL AMPLIFIERS AND OSCILLATORS							25ESC131/231.3 25ESC131/231.6			8 Hours		
Introduction to Op-Amps, Op-Amp Parameters, Inverting and Non-inverting Amplifiers. Oscillators: Positive Feedback, Condition for Oscillations, Wein Bridge Oscillator, RC phase shift oscillator.													
Case Study		Applications of Op-amp											
Text Book		Textbook 1: 12.1, 12.2, 12.4, 16.1, 16.2, 16.3											
MODULE-4	COMMUNICATION SYSTEMS							25ESC131/231.4 25ESC131/231.6			8 Hours		

Elements of Communication system, Need for modulation, types of communication systems, Analog modulation: Amplitude modulation, Frequency modulation, and Phase modulation (Definition and waveforms only), Digital modulation: ASK, FSK, PSK (Definition and waveforms only), Super heterodyne receiver.					
Case study	Converting an analog signal to a digital signal.				
Text Book	Textbook 2: 1.2, 1.3, 3.1, 4.1.1, 4.1.2, 6.2, 7.3.2				
MODULE-5	FUNDAMENTALS OF DIGITAL LOGIC	25ESC131/231.5 25ESC131/231.6	8 Hours		
Number Systems, Conversion from one number system to another, 1's and 2's Complement, Binary subtraction using 1's and 2's complement. Digital Electronics: Logic gates, NAND and NOR as universal gates, Boolean Algebra Theorems, De Morgan's theorem, Boolean algebraic simplification.					
Case study	4-bit adder simulator				
Text Book	Textbook 3: 1.2, 1.3, 1.4, 1.5, 2.1, 2.2, 2.3, 2.4, 2.7				
CIE Assessment Pattern (50 Marks – Theory)					
RBT Levels		Marks Distribution			
		Test (s)	AAT1	AAT2	AAT3
		25	7.5	7.5	10
L1	Remember	10	2.5	2.5	-
L2	Understand	10	5	-	-
L3	Apply	5	-	5	-
L4	Analyze	-	-	-	5
L5	Evaluate	-	-	-	-
L6	Create	-	-	-	5
SEE Assessment Pattern (50 Marks – Theory)					
RBT Levels		Exam Marks Distribution (50)			
L1	Remember	20			
L2	Understand	20			
L3	Apply	10			
L4	Analyze	-			
L5	Evaluate	-			
L6	Create	-			
Suggested Learning Resources:					
Text Books:					
1. Electronic Devices, Floyd, Thomas L.,10th edition, 2021, Pearson Education, Inc. ISBN-13: 978-0-13-254985-1					
2. Electronic Communication Systems, George Kennedy, 5 th edition, 2008, McGraw HILL Education Private Limited, ISBN:978-0-07-107782-8.					
3. Digital Logic and Computer Design, M. Morris Mano, 5 th edition, 2002, PHI, ISBN:978-0131989269.					
Reference Books:					
1. Digital Electronics: Principles, Devices and Applications, Anil K Maini, British Library Cataloguing in Publication Data, ISBN 978-0-470-03214-5 (HB)					
2. Principles of Electronics, V. K. Mehta, 12 th edition, 2020, S. Chand Publishing, ISBN:978-9352838277					
Web links and Video Lectures (e-Resources):					
• Basic Electronics: https://nptel.ac.in/courses/122106025					
• Embedded system: https://nptel.ac.in/courses/108102045					
• https://www.youtube.com/watch?vHbGxhPG3H5BQ					
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning					
➤ Hands-On Component Identification					
➤ Group Projects					
➤ Group activity to diagnose fault finding in basic simple electronic circuits.					

MODULE-1

Introduction to materials used in electronics, P-N Junction diode – operation, characteristics, Applications: Half-Wave rectifier, Center-tapped Full Wave Rectifier, Bridge Rectifier, Power supply filters and regulators.

MODULE 1

Introduction to materials used in electronics

Materials used in electronics

1. Conductors
2. Insulators
3. Semiconductors

Conductors- A conductor is a material that easily conducts electrical current. Most metals are good conductors, which are characterized by atoms with only one valence electron very loosely bound to the atom. These loosely bound valence electrons become free electrons. Therefore, in a conductive material the free electrons are valence electrons.

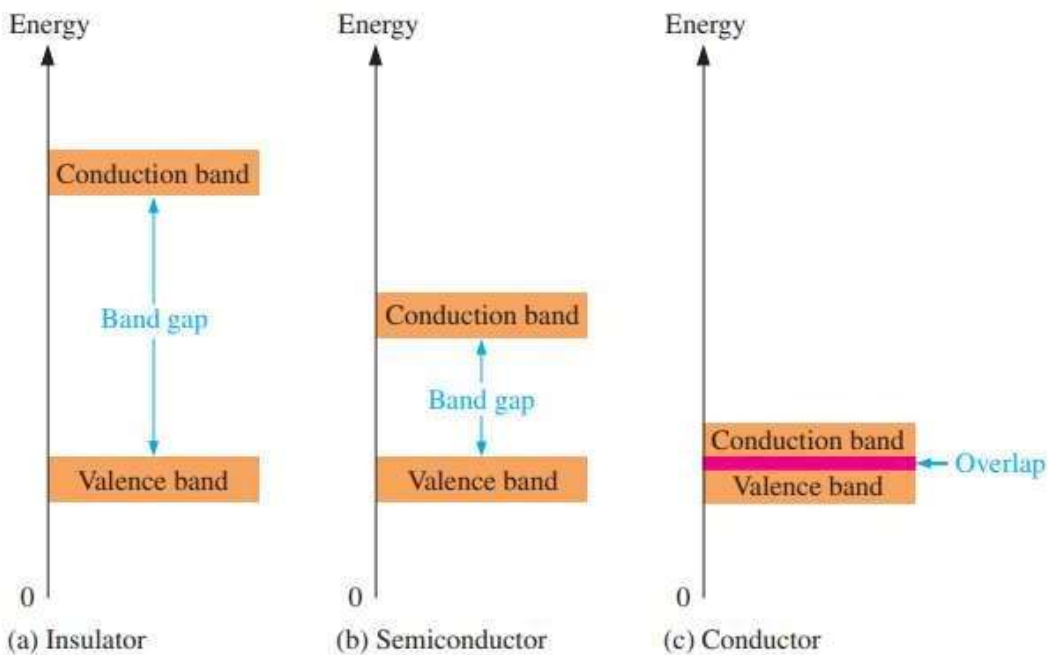
Examples of conductors are copper (Cu), silver (Ag), gold (Au), and aluminum (Al)

Insulators- An insulator is a material that does not conduct electrical current under normal conditions. □
Most good insulators are compounds rather than single-element materials and have very high resistivity. □
Valence electrons are tightly bound to the atoms; therefore, there are very few free electrons in an insulator.

Examples of insulators are rubber, plastics, glass, and quartz.

Semiconductors- A semiconductor is a material that is between conductors and insulators in its ability to conduct electrical current.

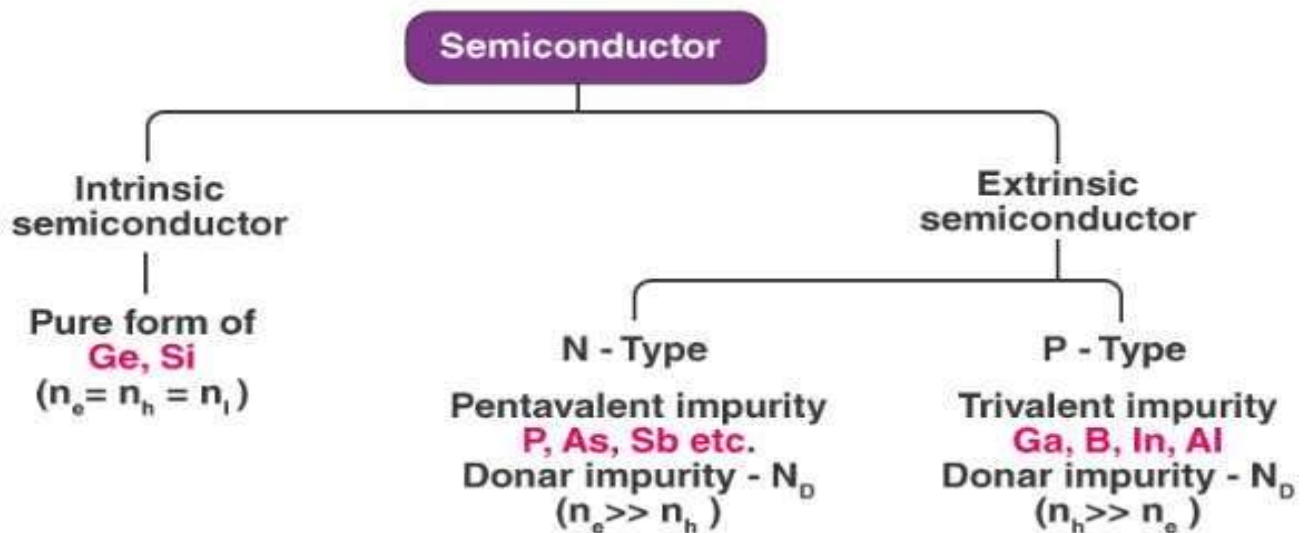
Examples of semiconductors are antimony (Sb), arsenic (As), boron (B), silicon (Si), and germanium (Ge).



Types of Semiconductors

Semiconductors can be classified as follows:

- Intrinsic Semiconductor
- Extrinsic Semiconductor



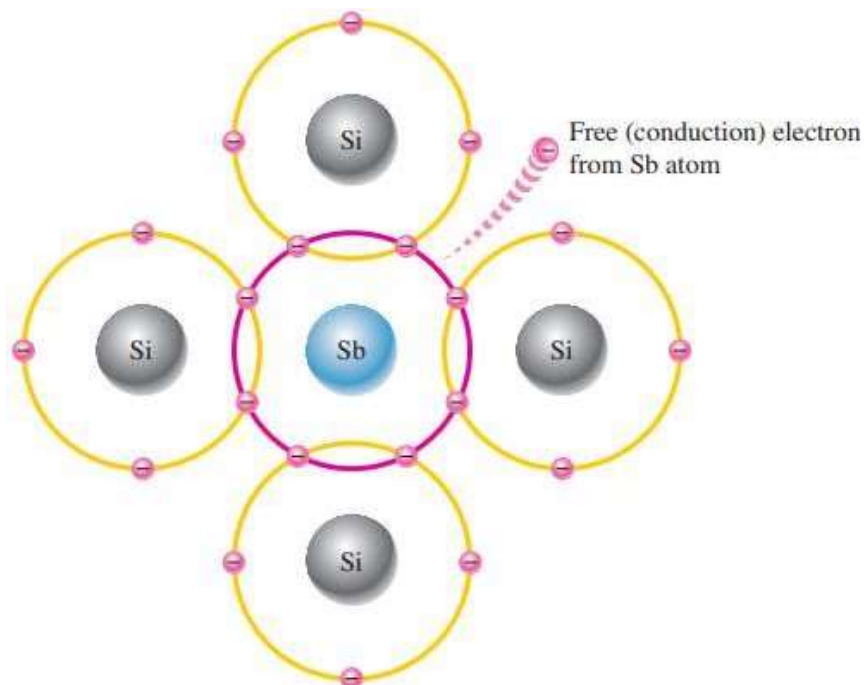
Classification of Semiconductors

Since semiconductors are generally poor conductors, their conductivity can be drastically increased by the controlled addition of impurities to the intrinsic (pure) semiconductive material. This process, called **doping**, increases the number of current carriers (electrons or holes). The two categories of impurities are n-type and p-type.

N-Type Semiconductor

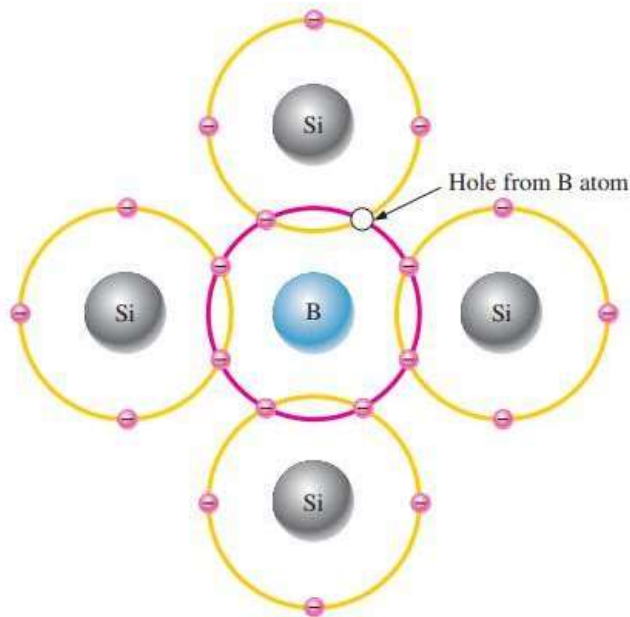
To increase the number of conduction-band electrons in intrinsic silicon, pentavalent impurity atoms are added. These are atoms with five valence electrons such as arsenic (As), phosphorus (P), bismuth (Bi), and antimony (Sb).

each pentavalent atom (antimony, in this case) forms covalent bonds with four adjacent silicon atoms. Four of the antimony atom's valence electrons are used to form the covalent bonds with silicon atoms, leaving one extra electron. This extra electron becomes a conduction electron because it is not involved in bonding. Because the pentavalent atom gives up an electron, it is often called a donor atom.



The electrons are called the majority carriers and Holes are minority charge carriers

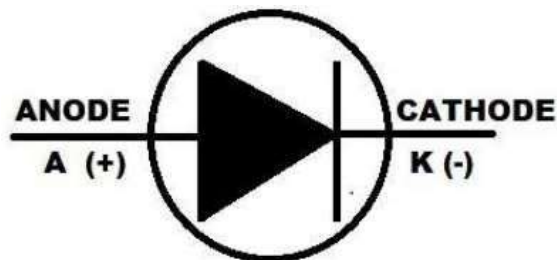
P-Type Semiconductor To increase the number of holes in intrinsic silicon, trivalent impurity atoms are added. These are atoms with three valence electrons such as boron (B), indium (In), and gallium (Ga). As illustrated in fig, each trivalent atom (boron, in this case) forms covalent bonds with four adjacent silicon atoms. All three of the boron atom's valence electrons are used in the covalent bonds; and, since four electrons are required, a hole results when each trivalent atom is added. Because the trivalent atom can take an electron, it is often referred to as an **acceptor atom**.



The Holes are called the majority carriers and Electrons are minority charge carriers

P-N Junction diode – operation, characteristics

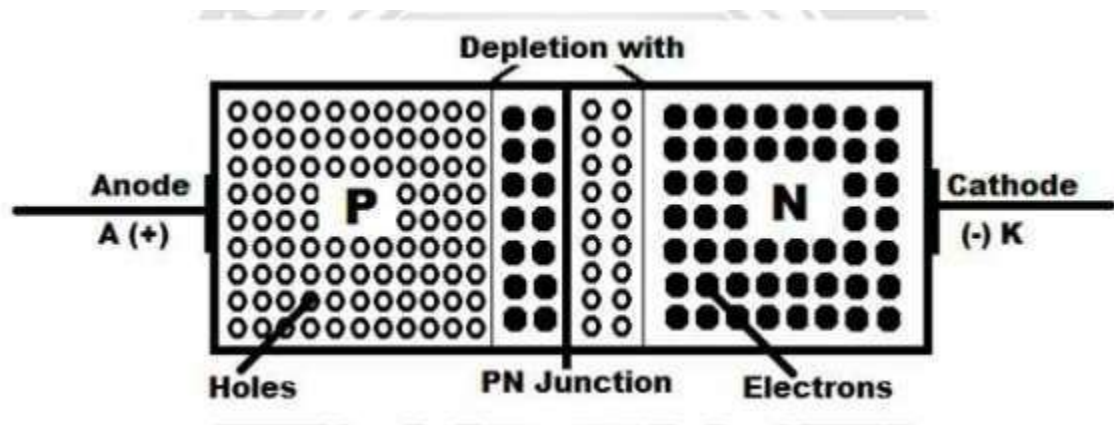
PN JUNCTION DIODE A diode is a device which only allows unidirectional flow of current if operated within a rated specified voltage level. A diode only blocks current in the reverse direction while the reverse voltage is within a limited range otherwise reverse barrier breaks and the voltage at which this breakdown occurs is called reverse breakdown voltage. The diode acts as a valve in the electronic and electrical circuit. A P-N junction is the simplest form of the diode which behaves as ideally short circuit when it is in forward biased and behaves as ideally open circuit when it is in the reverse biased. **SYMBOL OF DIODE** The name diode is derived from "di-ode" which means a device having two electrodes.



A simple PN junction diode is formed by doping donor impurity in one portion and acceptor impurity in other portion of silicon or germanium crystal block. This creates a p-n junction at the middle part of the block, where one portion is p-type (doped with trivalent or acceptor impurity), and another portion is n-type (doped with pentavalent or donor impurity). It can also be formed by joining a p-type (semiconductor doped with a

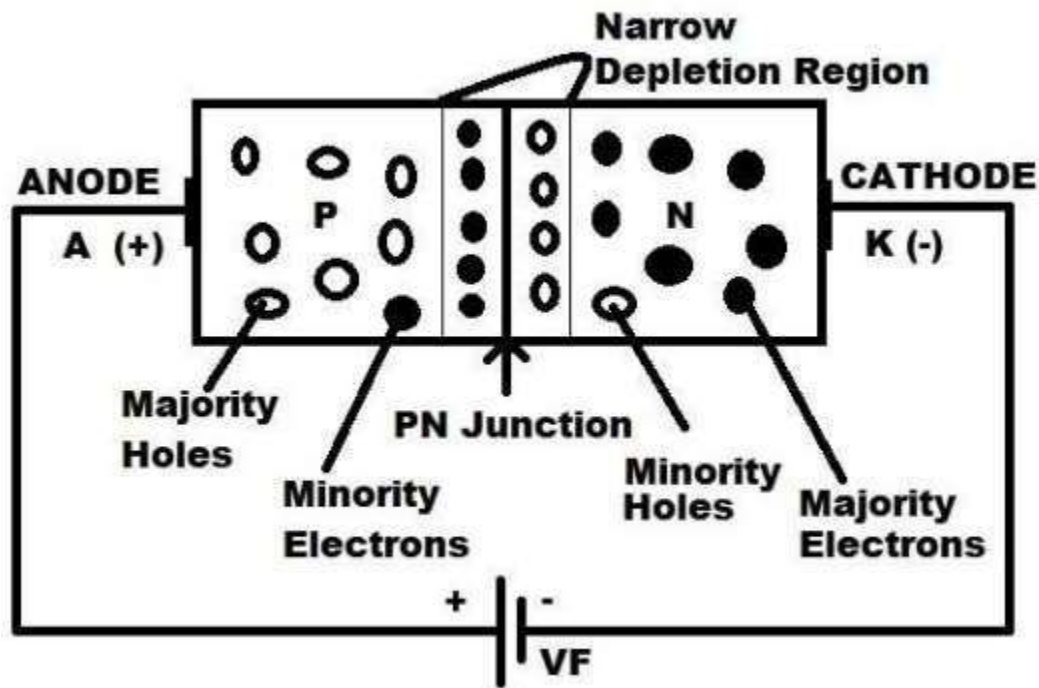
trivalent impurity) and n-type semiconductor (intrinsic semiconductor doped with a pentavalent impurity) together with a special fabrication technique such that a p-n junction is formed. It can also be formed by joining a p-type (semiconductor doped with a trivalent impurity) and n-type semiconductor (intrinsic semiconductor doped with a pentavalent impurity) together with a special fabrication technique such that a p-n junction is formed.

CONSTRUCTION The p-type forms anode and the n-type forms the cathode. These terminals are brought out to make the external connections. N-side will have a significant number of electrons, and very few holes (due to thermal excitation) whereas the p side will have a high concentration of holes and very few electrons. Due to this, a process called diffusion takes place. In this process free electrons from n side will diffuse (spread) into the p side and recombine with holes present there, leaving positive immobile (not moveable) ions in n side and creating negative immobile ions in p side of the diode. Hence, there will be uncovered positive donor ions in n-type side near the junction edge.

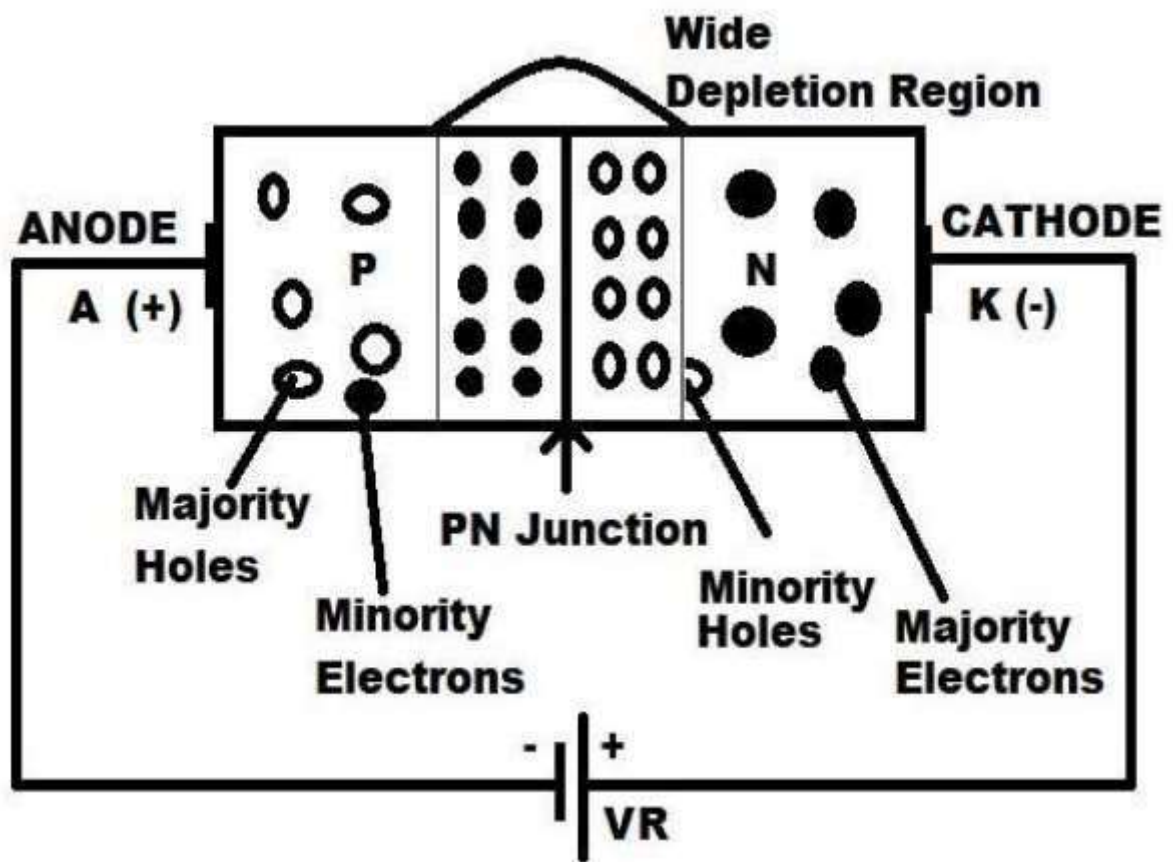


Similarly, there will be uncovered negative acceptor ions in p-type side near the junction edge. Due to this, numbers of positive ions and negative ions will accumulate on n-side and p-side respectively. This region so formed is called as depletion region due to the “depletion” of free carriers in the region. Due to the presence of these positive and negative ions a static electric field called as barrier potential is created across the pn junction of the diode. It is called as "barrier potential" because it acts as a barrier and opposes the further migration of holes and electrons across the junction. **FORWARD BIAS** In a PN junction diode when the forward voltage is applied i.e. positive terminal of a source is connected to the p-type side, and the negative terminal of the source is connected to the n-type side, the diode is said to be in forward biased condition.

There is a barrier potential across the junction. This barrier potential is directed in the opposite of the forward applied voltage. So a diode can only allow current to flow in the forward direction when forward applied voltage is more than barrier potential of the junction. This voltage is called forward biased voltage. For silicon diode, it is 0.7 volts. For germanium diode, it is 0.3 volts.



When forward applied voltage is more than this forward biased voltage, there will be forward current in the diode, and the diode will become short circuited. Hence, there will be no more voltage drop across the diode beyond this forward biased voltage, and forward current is only limited by the external resistance connected in series with the diode. Thus, if forward applied voltage increases from zero, the diode will start conducting only after this voltage reaches just above the barrier potential or forward biased voltage of the junction. The time, taken by this input voltage to reach that value or in other words, the time, taken by this input voltage to overcome the forward biased voltage is called recovery time. **REVERSE BIAS** The diode is reverse biased i.e. positive terminal of the source is connected to the n-type end, and the negative terminal of the source is connected to the p-type end of the diode, there will be no current through the diode except reverse saturation current. This is because at the reverse biased condition the depletion layer of the junction becomes wider with increasing reverse biased voltage. Although there is a tiny current flowing

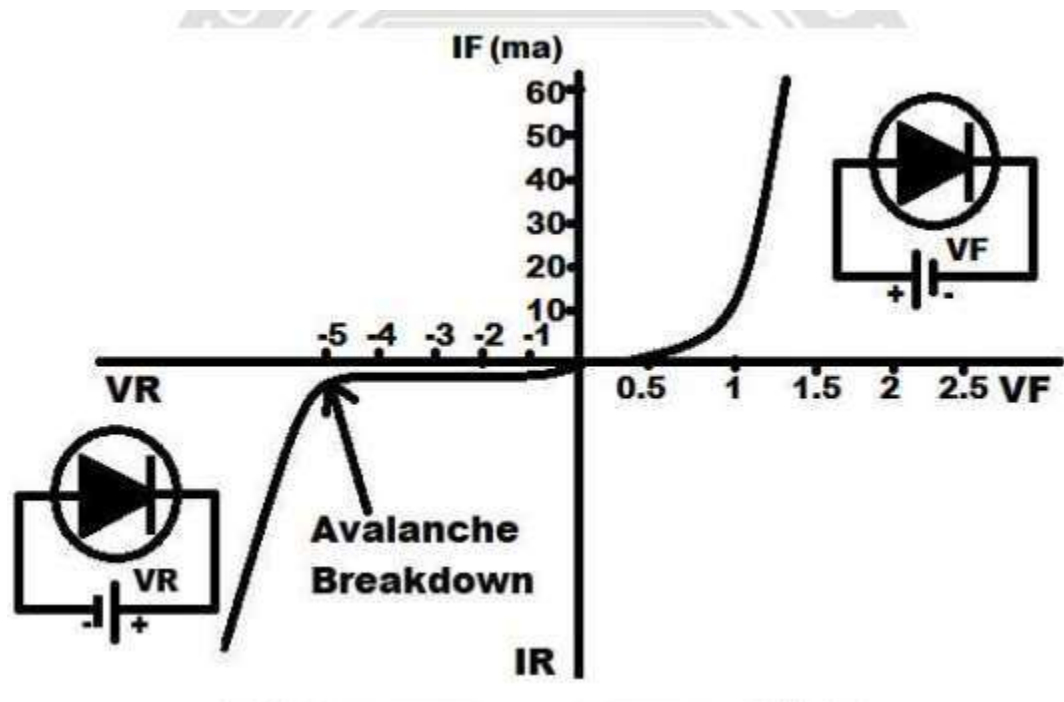


from n-type end to p-type end in the diode due to minority carriers. This tiny current is called reverse saturation current. Minority carriers are mainly thermally generated electrons and holes in p-type semiconductor and n-type semiconductor respectively.

Now if reverse applied voltage across the diode is continually increased, then after certain applied voltage the depletion layer will destroy which will cause a huge reverse current to flow through the diode. If this current is not externally limited and it reaches beyond the safe value, the diode may be permanently destroyed. This is because, as the magnitude of the reverse voltage increases, the kinetic energy of the minority charge carriers also increases. These fast moving electrons collide with the other atoms in the device to knock-off some more electrons from them. The electrons so released further release much more electrons from the atoms by breaking the covalent bonds. This process is termed as carrier multiplication and leads to a considerable increase in the flow of current through the p-n junction. The associated phenomenon is called Avalanche Breakdown.

V- I Characteristics

Forward Bias When, P terminal is more positive as compared to N-terminal i.e. P-terminal connected to positive terminal of battery and N-terminal connected to negative terminal of battery, it is said to be forward biased. Positive terminal of the battery repels majority carriers, holes, in P-region and negative terminal repels electrons in the N-region and push them towards the junction. This result in increase in concentration of carriers near junction, recombination takes place and width of depletion region decreases. As forward bias voltage is raised depletion region continues to reduce in width, and more and more carriers recombine. This results in exponential rise of current.



Reverse Bias In reverse biasing P- terminal is connected to negative terminal of the battery and N- terminal to positive terminal of battery. Thus applied voltage makes N-side more positive than P-side. Negative terminal of the battery attracts majority carriers, holes, in P-region and positive terminal attracts electrons in the N-region and pull them away from the junction. This result in decrease in concentration of charge carriers near junction and width of depletion region increases. A small amount of current flow due to minority carriers, called as reverse bias current or leakage current. As reverse bias voltage is raised

depletion region continues to increase in width and no current flows. It can be concluded that diode acts only when forward biased. Operation of diode can be summarized in form of I-V diode characteristics graph. For reverse bias diode,

$$V < 0, I_D = I_S$$

Where, V = supply voltage ID = diode current IS = reverse saturation current

$$V > 0, I_D = I_S(e^{V/NV_T} - 1)$$

For forward bias, Where, VT = volt's equivalent of temperature = KT/Q = T/11600 Q = electronic charge K = Boltzmann's constant N = 1, for Ge = 2, for Si

Applications

PN junction diodes have a wide range of applications, including rectifiers for AC to DC conversion, LEDs for lighting, photodiodes and solar cells for light detection and energy conversion, and use in voltage regulators, switches, oscillators, and various wave-shaping circuits like clippers and clamps.

Half-Wave rectifier

Alternating current (AC) Alternating current (AC) describes the flow of charge that changes direction periodically. As a result, the voltage level also reverses along with the current. AC is used to deliver power to houses, office buildings, etc.

The AC voltage of a periodic waveform may be written as

$$V(t) = V_m \sin(\omega t), (1)$$

where $\omega = 2\pi/T$ is the angular frequency of the waveform or voltage and T is the time period of the voltage.

Rectifier

Rectifier :-

Rectifier is a device that converts ac (alternating current) into dc (direct current). Semiconductor diodes are used as rectifying elements.

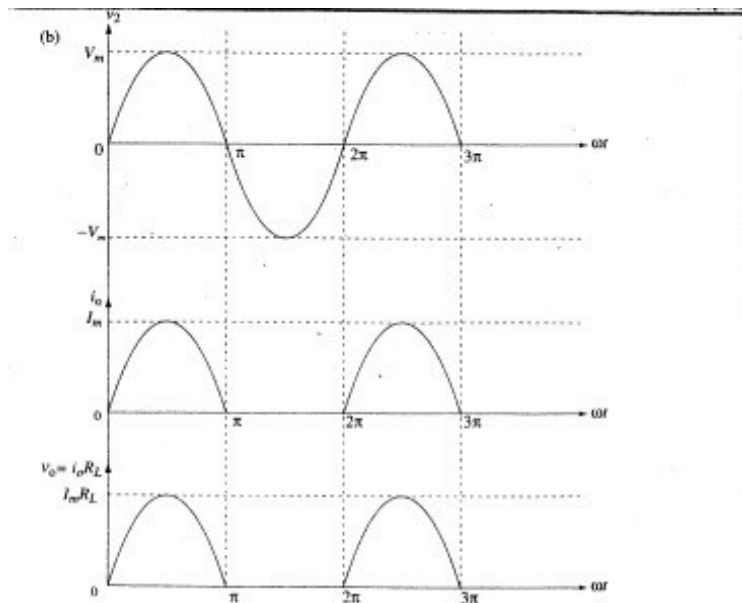
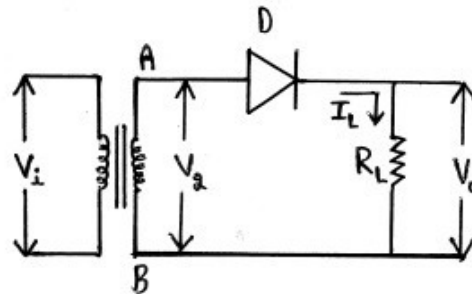
Rectifiers are classified into :

1. **Half wave rectifier (HWR)**
2. **Full wave rectifier (FWR)**

Full wave rectifier can be built in two ways :

1. **Full – wave rectifier using two diodes and a centre tapped transformer**
2. **Full-wave bridge rectifier using four diodes and an ordinary transformer.**

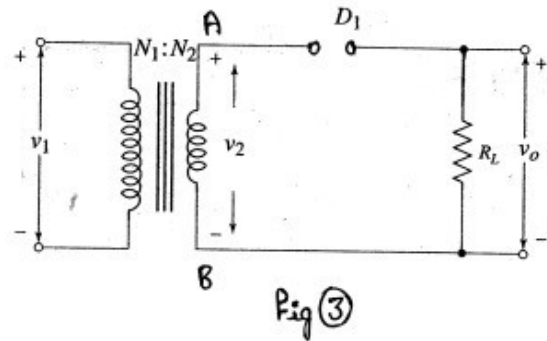
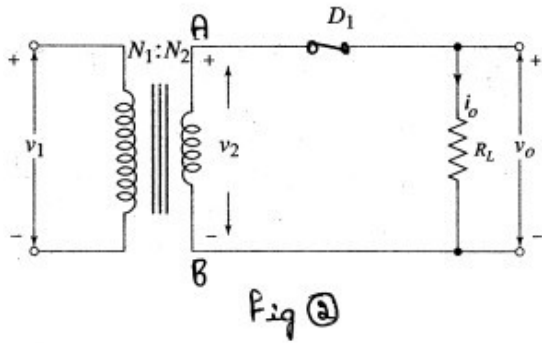
HALFWAVE RECTIFIER (HWR):-



a) Half-wave rectifier. (b) Waveforms of transformer secondary voltage, load current and load voltage.

Operation :-

During positive half cycle of the ac input voltage, end A becomes positive with respect to end B, the diode 'D' is forward biased and acts as a short circuit, thus the current flows in the circuit, Figure 2. The load voltage is given by $V_0 = I_L R_L$.



During negative half cycle of the ac input voltage, end A becomes negative with respect to end B, the diode 'D' is reverse biased and acts as an open circuit, thus **NO current** flows in the circuit, Figure 3. The load voltage is given by $V_0 = 0 \times R_L$.

Therefore

$$V_0 = 0v$$

$$I_{dc} = \frac{I_m}{\pi}$$

$$I_{RMS} = \frac{I_m}{2}$$

DC output power (P_{dc}) :-

$$\begin{aligned} P_{dc} &= I_{dc}^2 R_L \\ &= \left(\frac{I_m}{\pi} \right)^2 R_L = \frac{I_m^2}{\pi^2} R_L \end{aligned}$$

AC output power (P_{ac}) :-

$$P_{ac} = I_{RMS}^2 R_L$$

$$P_{ac} = \left(\frac{I_m}{2} \right)^2 R_L$$

Rectification efficiency (η):-

The rectifier efficiency is defined as the ratio of output dc power to input ac power.

$$\eta = \frac{\text{DC o/p power}}{\text{AC I/p power}} = \frac{P_{dc}}{P_{ac}} = \frac{\frac{I_m^2}{\pi^2} R_L}{\frac{I_m^2}{4} R_L} = \frac{\frac{1}{\pi^2}}{\frac{1}{4}} = \frac{1}{\pi^2} \times \frac{4}{1}$$

$$\eta = 0.406$$

$$\boxed{\% \eta = 40.6\%}$$

In HWR, maximum 40.6% ac power gets converted to dc power in the load.

Ripple factor (γ):-

Ripple factor is defined as the ratio of RMS value of the ac component present in the rectified output to the dc component present in the rectified output.

$$\gamma = \frac{\text{RMS value of ac Component of o/p}}{\text{dc Component of o/p}}$$

$$\boxed{\gamma = \frac{I_{ac}}{I_{dc}}} \rightarrow \textcircled{1}$$

$$\text{WKT } I_{RMS} = \sqrt{I_{ac}^2 + I_{dc}^2}$$

$$I_{RMS}^2 = I_{ac}^2 + I_{dc}^2$$

$$I_{ac}^2 = I_{RMS}^2 - I_{dc}^2$$

$$\boxed{I_{ac} = \sqrt{I_{RMS}^2 - I_{dc}^2}} \rightarrow \textcircled{2}$$

WKT $\boxed{I_{dc} = \frac{I_m}{\pi}} \rightarrow (3)$

Substituting eq (2) & (3) in eq (1), we get

$$\gamma = \frac{I_{ac}}{I_{dc}} = \frac{\sqrt{I_{RMS}^2 - I_{dc}^2}}{I_{dc}} = \sqrt{\frac{I_{RMS}^2 - I_{dc}^2}{I_{dc}^2}} = \sqrt{\frac{I_{RMS}^2}{I_{dc}^2} - \frac{I_{dc}^2}{I_{dc}^2}}$$

$$= \sqrt{\frac{I_{RMS}^2}{I_{dc}^2} - 1} = \sqrt{\frac{(\frac{I_m}{2})^2}{(\frac{I_m}{\pi})^2} - 1}$$

$$= \sqrt{\frac{\frac{1}{4}}{\frac{1}{\pi^2}} - 1} = \sqrt{\frac{\pi^2}{4} - 1}$$

$\boxed{\gamma = 1.21}$

This indicates that the **amount of ac present in the output is 121 % of the dc voltage.**

FWR with centre-tapped transformer :-

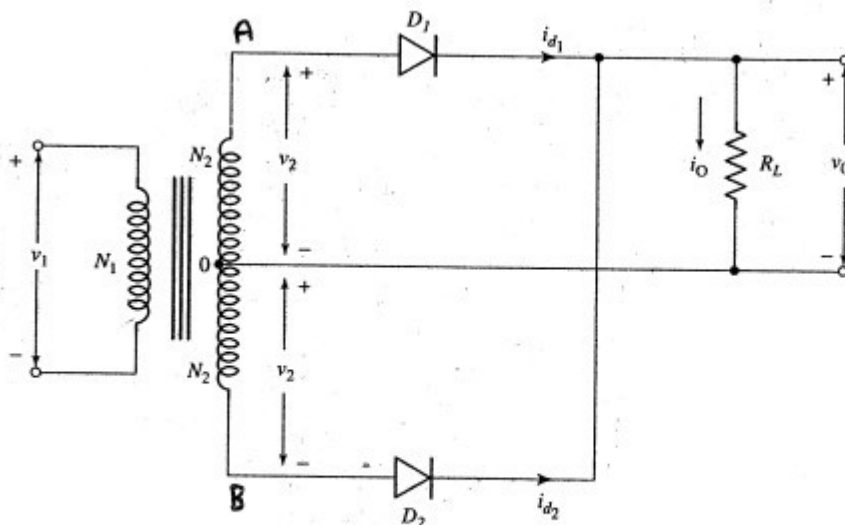


Figure 1. FWR using two diodes & a centre-tapped transformer.

Operation :-

During +ve half cycle :-

❖ During +ve half cycle of the ac input voltage, end A becomes +ve with respect to end B, the diode ' D_1 ' is forward biased and conducts while the diode D_2 is reverse biased and acts as open circuit and will not conduct as shown in fig 2.

❖ The diode **D_1** supplies the **load current**. The conventional **current flow** is through diode **D_1** , load resistor **R_L** & the **upper half** of **secondary winding** as shown by the **dotted arrows**.

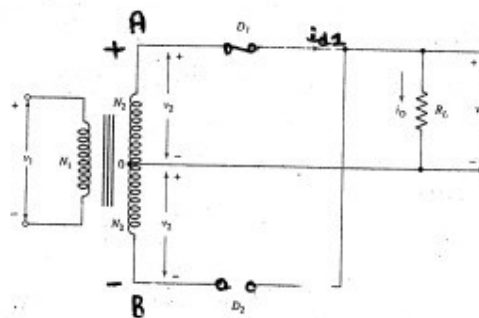


Fig 2.

During -ve half cycle :-

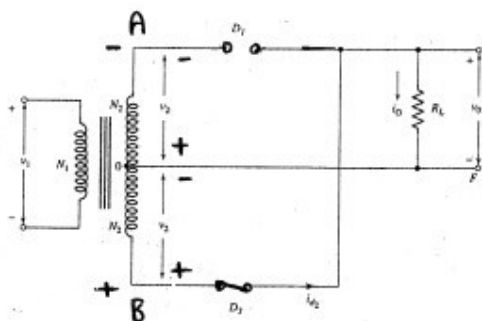


Fig 3.

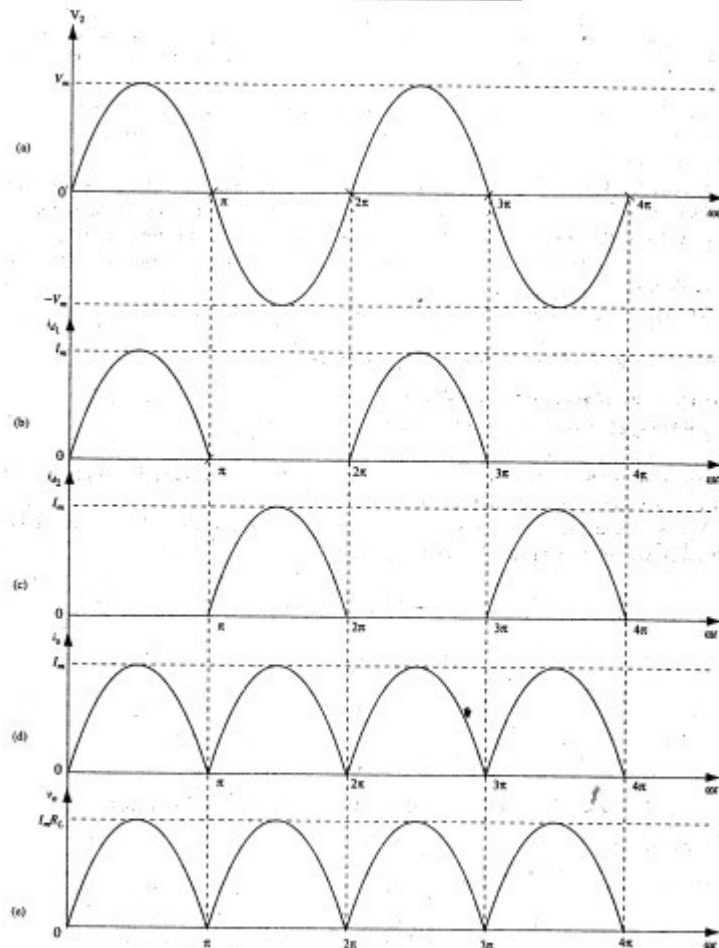
❖ During -ve half cycle of the ac input voltage, end A becomes -ve with respect to end B, the diode ' D_2 ' is forward biased and conducts while the diode D_1 is reverse biased and acts as open circuit and will not conduct as shown in fig 3.

❖ The diode **D_2** supplies the **load current**. The conventional **current flow** is through diode **D_2** , load resistor **R_L** & the **lower half** of **secondary winding** as shown by the **solid arrows**.

❖ From fig 2 & 3 it can be observed that **current** in the **load R_L** is in the **same direction** for **both half cycles** of **ac input voltage**.

❖ For both the half cycles the current flows through load in the same direction. Hence we get two half cycles for one complete input signal.

I/O Waveforms



Voltage and current waveforms. (a) Secondary voltage waveform, (b) & (c) Diode current waveforms, (d) load current waveform, (e) load voltage waveform.

$$I_{dc} = \frac{2I_m}{\pi}$$

$$I_{RMS} = \frac{I_m}{\sqrt{2}}$$

DC output power (P_{dc}) :-

$$P_{dc} = I_{dc}^2 R_L$$

$$= \left(\frac{2I_m}{\pi} \right)^2 R_L$$

$$P_{dc} = \frac{4}{\pi^2} I_m^2 R_L$$

AC output power (P_{ac})

$$P_{ac} = I_{RMS}^2 R_L$$

$$= \left(\frac{I_m}{\sqrt{2}} \right)^2 R_L$$

$$P_{ac} = \frac{I_m^2}{2} R_L$$

Ripple factor (γ):-

$$\gamma = \sqrt{\frac{I_{RMS}^2}{I_{dc}^2} - 1}$$

$$\gamma = \sqrt{\left[\frac{\left(\frac{I_m}{\sqrt{2}}\right)^2}{\left(\frac{2I_m}{\pi}\right)^2} \right] - 1}$$

$$= \sqrt{\left[\frac{\frac{I_m^2}{2}}{\frac{4I_m^2}{\pi^2}} \right] - 1} = \sqrt{\left(\frac{\frac{1}{2}}{\frac{4}{\pi^2}} \right) - 1}$$

$$= \sqrt{\left(\frac{1}{2} \times \frac{\pi^2}{4} \right) - 1}$$

$$= \sqrt{\frac{\pi^2}{8} - 1}$$

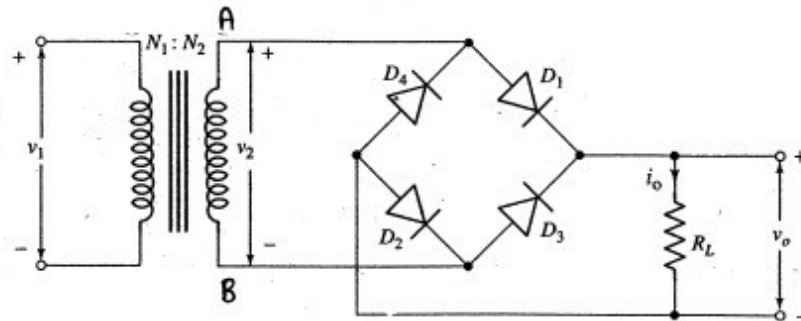
$$\boxed{\gamma = 0.48}$$

Rectification efficiency (η):-

$$\eta = \frac{P_{ac}}{P_{dc}} = \frac{\frac{4}{\pi^2} \frac{I_m^2}{2} R_L}{\frac{I_m^2}{2} R_L} = \frac{\frac{4}{\pi^2}}{\frac{1}{2}} = \frac{4}{\pi^2} \times \frac{2}{1} = 0.812$$

$$\boxed{\% \eta = 81.2\%}$$

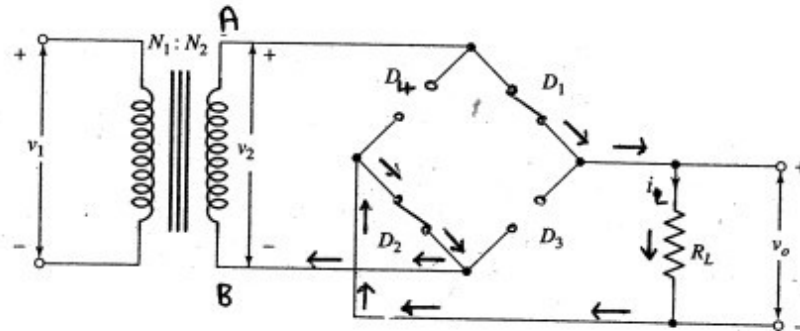
FULL WAVE BRIDGE Rectifier (FWBR) :-



Full-wave bridge rectifier.

Operation :-

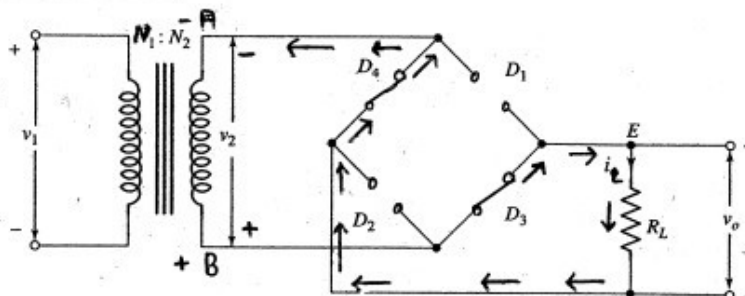
During +ve half cycle :-



- ❖ During **+ve half cycle** of the ac input voltage, end A becomes +ve with respect to end B. This makes diodes ***D1 & D2 forward biased***, while ***D3 & D4 are reverse biased***.

Therefore only **diodes D1 & D2 conducts**. The conventional **current flows** through the **load resistance R_L** and is shown by the **arrows**.

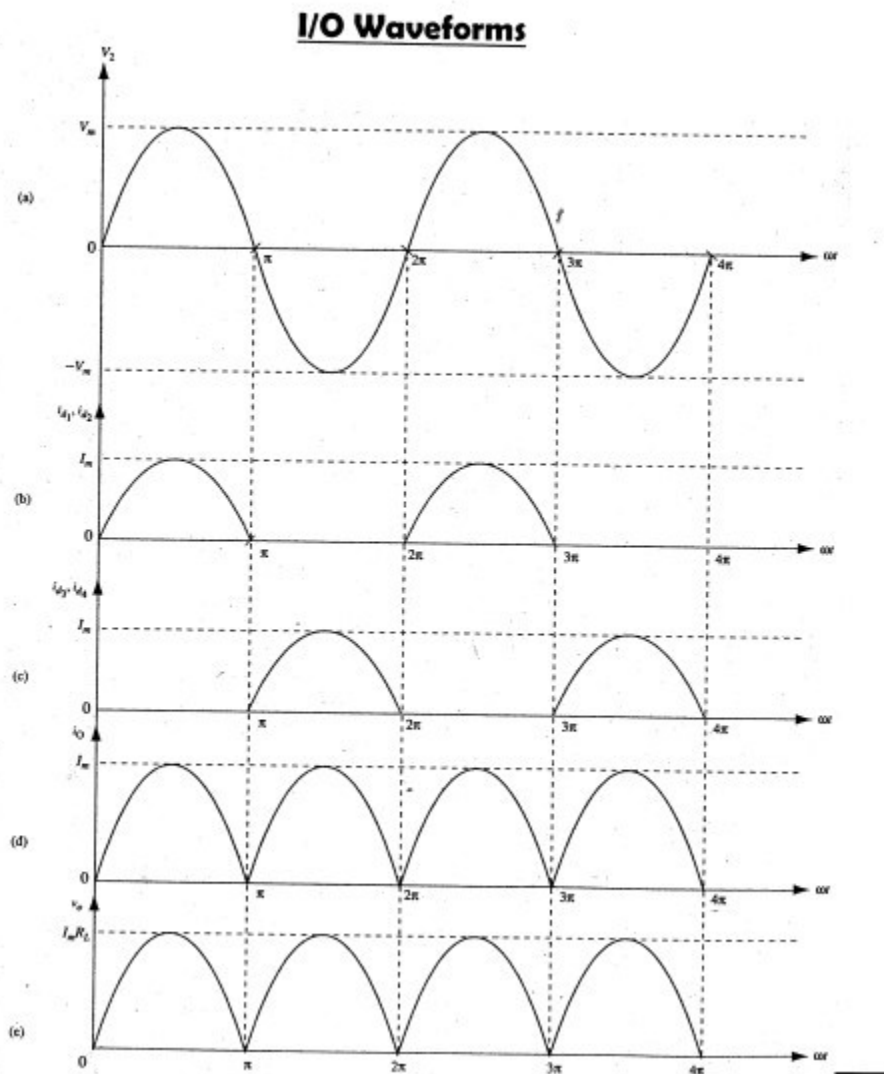
During -ve half cycle :-



- ❖ During **-ve half cycle** of the ac input voltage, end A becomes -ve with respect to end B. This makes diodes **D_3 & D_4 forward biased**, while **D_1 & D_2 are reverse biased**.

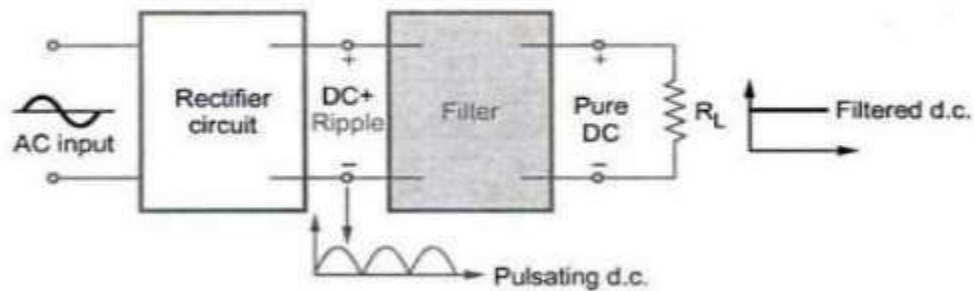
Therefore only **diodes D_3 & D_4 conducts**. The conventional **current flows** through the **load resistance R_L** and is shown by the **arrows**.

- ❖ For both the half cycles the current flows through load in the same direction. Hence we get two half cycles for one complete input signal.



Power Supply Filters

It is seen that the output a half-wave or full wave rectifier circuit is not pure d.c.; but it contains fluctuations or ripple, which are undesired. To minimize the ripple content in the output, filter circuits are used. These circuits are connected between the rectifier and load, as shown in the Fig. 1.13

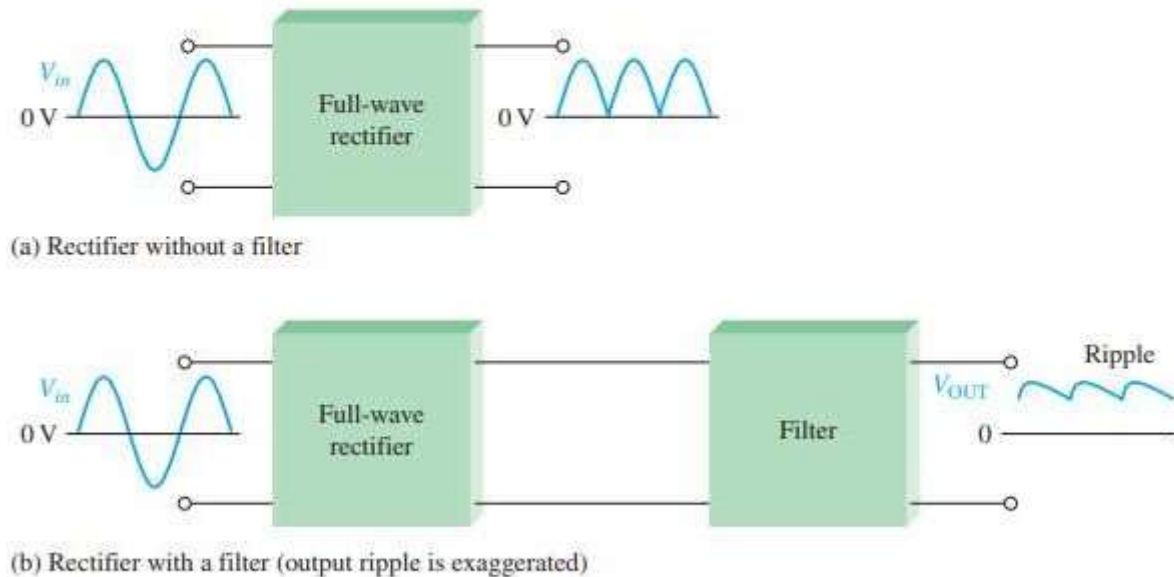


Power supply using rectifier and filter

Power Supply Filters

A power supply filter ideally eliminates the fluctuations in the output voltage of a halfwave or full-wave rectifier and produces a constant-level dc voltage. Filtering is necessary because electronic circuits require a constant source of dc voltage and current to provide power and biasing for proper operation. Filters are implemented with capacitors, as you will see in this section. Voltage regulation in power supplies is usually done with integrated circuit voltage regulators. A voltage regulator prevents changes in the filtered dc voltage due to variations in input voltage or load.

In most power supply applications, the standard 60 Hz ac power line voltage must be converted to an approximately constant dc voltage. The 60 Hz pulsating dc output of a half-wave rectifier or the 120 Hz pulsating output of a full-wave rectifier must be filtered to reduce the large voltage variations. Figure illustrates the filtering concept showing a nearly smooth dc output voltage from the filter. The small amount of fluctuation in the filter output voltage is called *ripple*.



▲ **FIGURE**

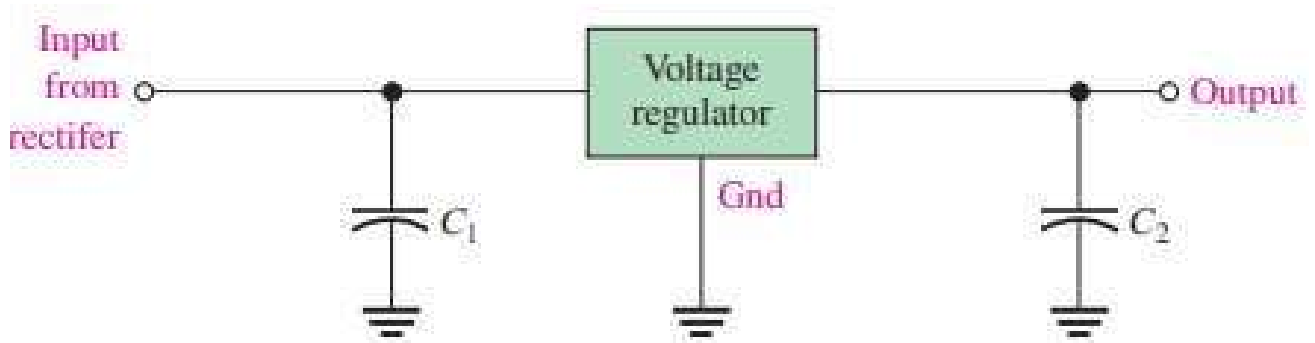
Power supply filtering.

Voltage Regulators

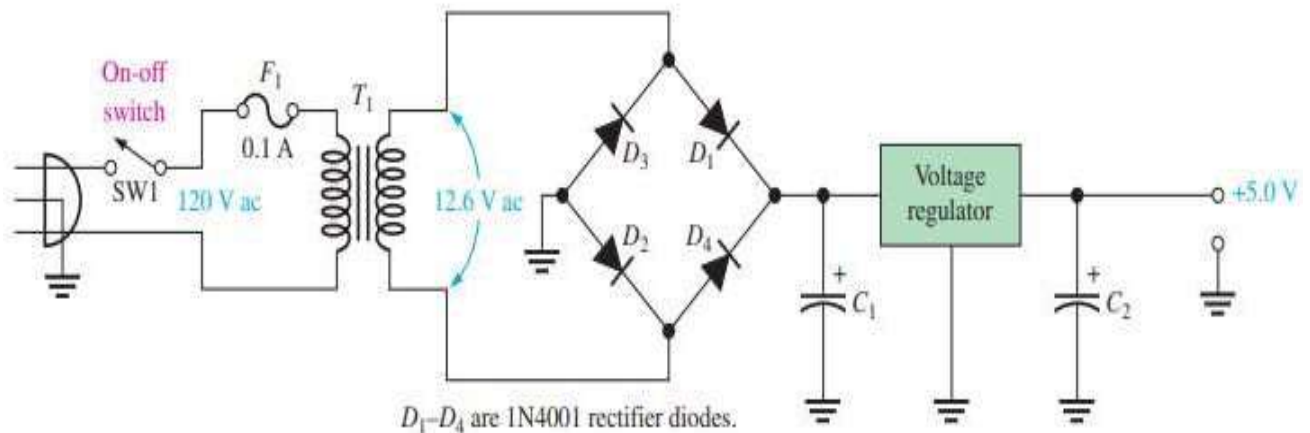
. A voltage regulator is connected to the output of a filtered rectifier and maintains a constant output voltage (or current) despite changes in the input, the load current, or the temperature. The capacitor-input filter reduces the input ripple to the regulator to an acceptable level. The combination of a large capacitor and a voltage regulator helps produce an excellent power supply.

Most regulators are integrated circuits and have three terminals—an input terminal, an output terminal, and a reference (or adjust) terminal. The input to the regulator is first filtered with a capacitor to reduce the ripple to <10%. The regulator reduces the ripple to a negligible amount. In addition, most regulators have an internal voltage reference, shortcircuit protection, and thermal shutdown circuitry. They are available in a variety of voltages, including positive and negative outputs, and can be designed for variable outputs with a minimum of external components. Typically, voltage regulators can furnish a constant output of one or more amps of current with high ripple rejection. Three-terminal regulators designed for fixed output voltages require only

external capacitors to complete the regulation portion of the power supply, as shown in Figure 2–50. Filtering is accomplished by a large-value capacitor between the input voltage and ground. An output capacitor (typically) is connected from the output to ground to improve the transient response. 0.1 μF to 1.0 μF



A basic fixed power supply with a +5 V voltage regulator is shown in Figure



A basic +5.0 V regulated power supply.

Percent Regulation

The regulation expressed as a percentage is a figure of merit used to specify the performance of a voltage regulator. It can be in terms of input (line) regulation or load regulation.

Line Regulation The **line regulation** specifies how much change occurs in the output voltage for a given change in the input voltage. It is typically defined as a ratio of a change in output voltage for a corresponding change in the input voltage expressed as a percentage.

$$\text{Line regulation} = \left(\frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{IN}}} \right) 100\%$$

Load Regulation The **load regulation** specifies how much change occurs in the output voltage over a certain range of load current values, usually from minimum current (no load, NL) to maximum current (full load, FL). It is normally expressed as a percentage and can be calculated with the following formula:

$$\text{Load regulation} = \left(\frac{V_{\text{NL}} - V_{\text{FL}}}{V_{\text{FL}}} \right) 100\%$$

where V_{NL} is the output voltage with no load and V_{FL} is the output voltage with full (maximum) load.